MAJOR PROJECT · SEMESTER 7

Strategic Application of Artificial Intelligence in CAD

A written requirement goes in. A sized, checked, modelled and drawn machine joint comes out, with every number traceable to a published formula.

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THE PROBLEM

The calculation is repetitive, and repeating it is where the time goes

For a single knuckle joint the designer computes the rod from tension, sizes the pin twice, derives seven proportions, then verifies nine separate failure modes. Change the load and every one of those steps is done again.

Parametric CAD removes the drafting labour but not this. A modeller will regenerate whatever dimensions it is given, and will not object if the pin is too slender to carry its bending moment.

> **The aim.** Keep every formula exactly as the textbook has it and remove only the repetition, so that the engineering stays checkable.

STEP DONE BY HAND, PER JOINT	REPEATS ON LOAD CHANGE
Rod diameter from direct tension	yes
Pin from double shear	yes
Pin from bending, take the larger	yes
Seven proportions from the rod	yes
Nine failure checks	yes
Re-size anything that fails, re-check	yes
Four views, dimensions, title block	yes

OBJECTIVES

What the project set out to do

- **Interpret a written requirement.** Extract joint type, load and allowable stresses from a sentence in ordinary English.
- **Size from published formulas.** Use the procedure given in the design literature, rounding to preferred sizes as the source directs.
- **Verify every failure mode** against the allowable stress, and iterate where a check is not satisfied.
- **Generate exact solid models** of each component and the assembly, in a format that opens in commercial CAD.
- **Produce a first angle drawing sheet** with dimensions and a title block, to Indian drawing practice.
- **Verify the system itself** against published design literature, and record any disagreement with a source explicitly rather than silently.
- **Make it usable in a browser** with nothing installed.

> **Measure of success.** Every dimension the system produces can be attributed either to a formula with a cited source, or to a named failure check. Nothing is unexplained.

SYSTEM OVERVIEW

Five stages, each reporting its result to the next

STAGE 1 · AI

Read the request

Sentence to parameters: joint, load, allowable stresses. Pattern rules as fallback.

STAGE 2

Size

Governing formula gives the primary diameter, rounded up to a preferred size. Proportions follow.

STAGE 3

Check

Every failure mode evaluated against its allowable. Failing dimension raised 1 mm and re-checked.

STAGE 4

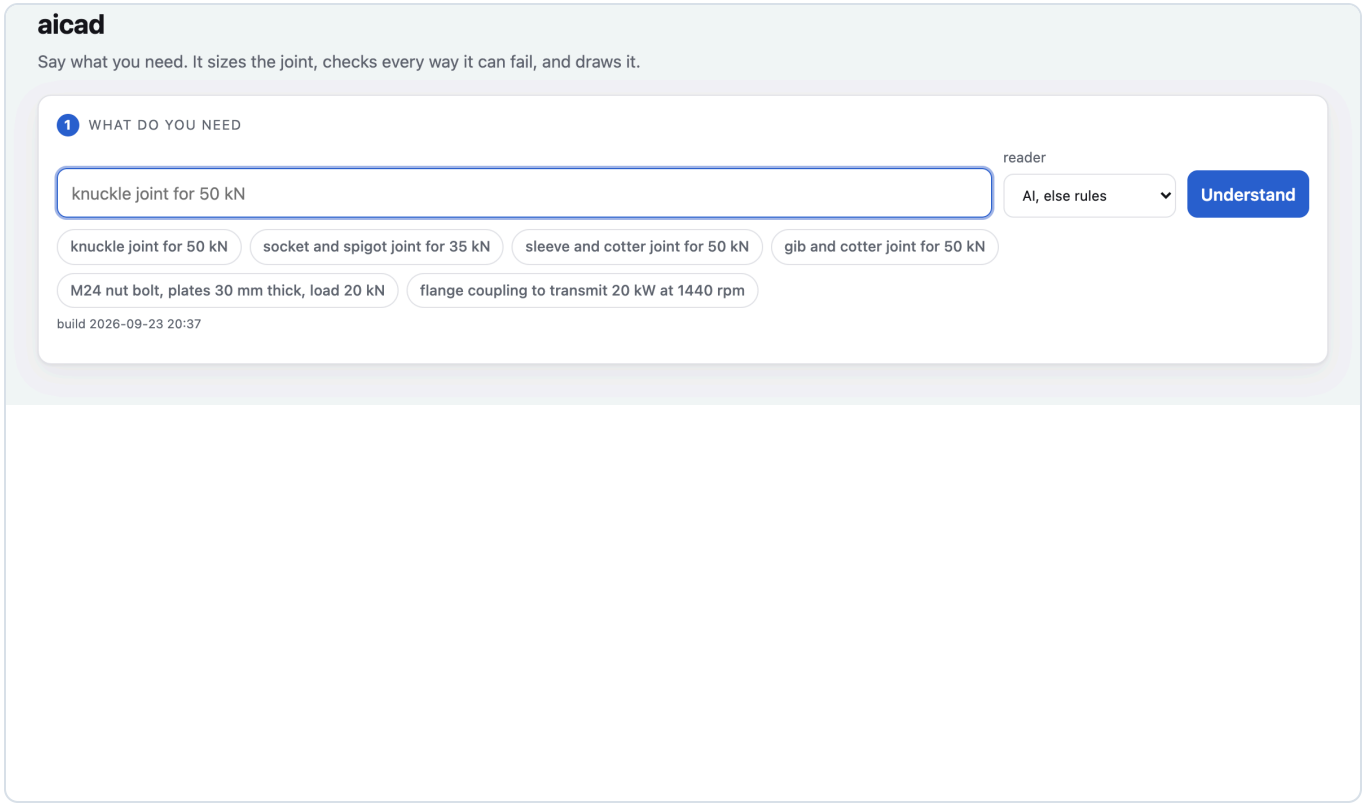
Model

Each part built as an exact solid, written as STEP, plus the assembly in position.

STAGE 5

Draw

First angle A3 sheet, four views, dimensions, title block, and a DXF.



The request screen. The design rules for each joint are held as data, not code, so the source of any dimension can be reported as it is computed.

6

JOINTS IMPLEMENTED

44

DISTINCT FAILURE CHECKS

290

AUTOMATED TESTS

THE CENTRAL DESIGN DECISION

The AI reads the request. It selects no dimension and performs no check.

Determining that `50 kN` denotes 50,000 N, and that *socket and spigot* names the cotter joint, is a language problem. A language model is effective at that.

Determining the pin diameter is an engineering problem with one correct answer obtainable from a formula. It stays with deterministic code whose every step is traceable to a printed page.

Safeguard. Any number the model returns that does not appear in the original request is discarded and the user is told. A plausible but invented figure cannot enter the calculation.

Why not let it generate the whole model? Because then no dimension could be checked against anything. Geometry that cannot be justified is of no use in a design review.

STAGE	PERFORMED BY	CHECKABLE AGAINST
Read the sentence	**Language model**	the sentence itself
Size every dimension	formula	cited book page
Verify failure modes	formula	cited book page
Build solids	CAD kernel	the computed dimensions
Draw the sheet	FreeCAD	the assembled geometry

SIZING PROCEDURE

How a dimension is arrived at

1. Governing dimension from the primary failure mode

$$D = \sqrt{\frac{4P}{\pi \sigma_t}}$$

The computed value is rounded **up to the next preferred size**, as the source directs, not used as computed.

2. A component with two independent modes is sized from both

$$\text{pin shear: } d = \sqrt{\frac{2P}{\pi \tau}}$$

$$\text{pin bending: } d = \left[\frac{16P\,(t_1/3 + t/4)}{\pi \sigma_b} \right]^{1/3}$$

The **larger requirement governs**. A pin sized from shear alone satisfies that check and bends in service.

3. Remaining dimensions from book proportions

Each expressed as a multiple of the governing dimension, with its source recorded beside it, for example

$$d_2 = 2.0\,D \qquad t = 1.25\,D \qquad t_1 = 0.75\,D$$

NPTEL IIT KGP, Module 4 Lesson 2, Fig. 4.2.3.1

4. Iterate until every check passes

Where a check fails, the responsible dimension is raised by 1 mm and the whole set is re-evaluated. Both the dimension raised and the check that caused it are reported.

> **Design factor 1.25.** The joint is sized for 1.25 times the stated load, then every stress is reported at the actual load. The governing check lands near 75 to 80 percent of allowable instead of 100, so margin is retained for material variation and overload.

SCOPE

Six joints, 44 failure checks in total

JOINT	PARTS	DIMENSIONS	CHECKS	DUTY USED HERE	BUSIEST	UTILISATION OF ALLOWABLE
Knuckle joint	5	12	9	Load P = 50,000 N	76%	
Socket and spigot cotter joint	3	14	10	Load P = 35,000 N	67%	
Sleeve and cotter joint	5	12	8	Load P = 50,000 N	77%	
Gib and cotter joint	4	20	8	Load P = 50,000 N	79%	
Hex bolt, nut and washer	3	11	2	Load P = 20,000 N	71%	
Protected type rigid flange coupling	12	15	7	Torque T = 500,000 N mm (500 N m)	79%	

Reading the last column. Every joint passes every check, and the busiest check sits between 67 and 79 percent of allowable. That spread is the design factor working: sizing exactly to the allowable would leave a design that passes on paper with nothing in reserve.

The gib and cotter joint needed a decision. Unwin gives the breadth split and the thickness but no strap proportions. Rather than borrow a ratio or invent one, every dimension the source does not give is sized against a failure check until it passes. No dimension of that joint is unsupported.

WORKED EXAMPLE · 1 OF 3

Knuckle joint for 50 kN: which formula governs

COMPUTED BEFORE ROUNDING	MM	EFFECT
Rod in direct tension	32.57	rod adopted as **35**
Pin in double shear	25.75	not governing
Pin in bending	43.72	governs, pin adopted as **45**

> **The point of this slide.** The bending requirement is nearly **1.7 times** the shear requirement. A pin sized from shear alone would satisfy its own check and bend in service. The source material takes the same step.

Allowable stresses used: 75 MPa tension, 60 MPa shear, 150 MPa crushing. Sized for 62,500 N, reported at 50,000 N.

SYMBOL	MEANING	MM	RULE AND SOURCE
D	rod diameter	35	rod in tension: D = sqrt(4P / (pi sigma_t)), rounded up to a preferred siz
d	pin diameter	45	governed by pin bending
d2	eye outer diameter	70	d2 = 2.0 x D, the rod diameter (outer diameter of eye) [NPTEL IIT KGP M4L2
d3	pin head and collar diameter	52.5	d3 = 1.5 x D, the rod diameter (pin head and collar diameter) [NPTEL IIT K
t	eye thickness	43.75	t = 1.25 x D, the rod diameter (thickness of single eye) [NPTEL IIT KGP M4
t1	fork plate thickness	26.25	t1 = 0.75 x D, the rod diameter (thickness of each fork plate) [NPTEL IIT
t2	pin head thickness	17.5	t2 = 0.5 x D, the rod diameter (thickness of pin head) [NPTEL IIT KGP M4L2
collar_t	collar thickness	22.5	-

WORKED EXAMPLE · 2 OF 3

All nine failure modes, evaluated at the actual load

FAILURE MODE	EXPRESSION	STRESS MPA	ALLOWABLE	USED	UTILISATION
rod tension	`4P / (pi D^2)`	52	75	69%	
pin double shear	`P / (2 pi/4 d^2)`	15.7	60	26%	
pin bending	`32M / (pi d^3), M = P/2 (t1/3 + t/4)`	55	75	73%	
eye tension	`P / ((d2 – d) t)`	45.7	75	61%	
eye shear	`P / ((d2 – d) t)`	45.7	60	76%	
eye crushing	`P / (d t)`	25.4	150	17%	
fork tension	`P / ((d2 – d) 2 t1)`	38.1	75	51%	
fork shear	`P / ((d2 – d) 2 t1)`	38.1	60	64%	
fork crushing	`P / (d 2 t1)`	21.2	150	14%	

Every check satisfied. The busiest is shear of the eye at 76 percent of its allowable stress.

Nothing here is stored. These are computed for this request. At 80 kN every value in the table changes, and the pin would be re-sized.

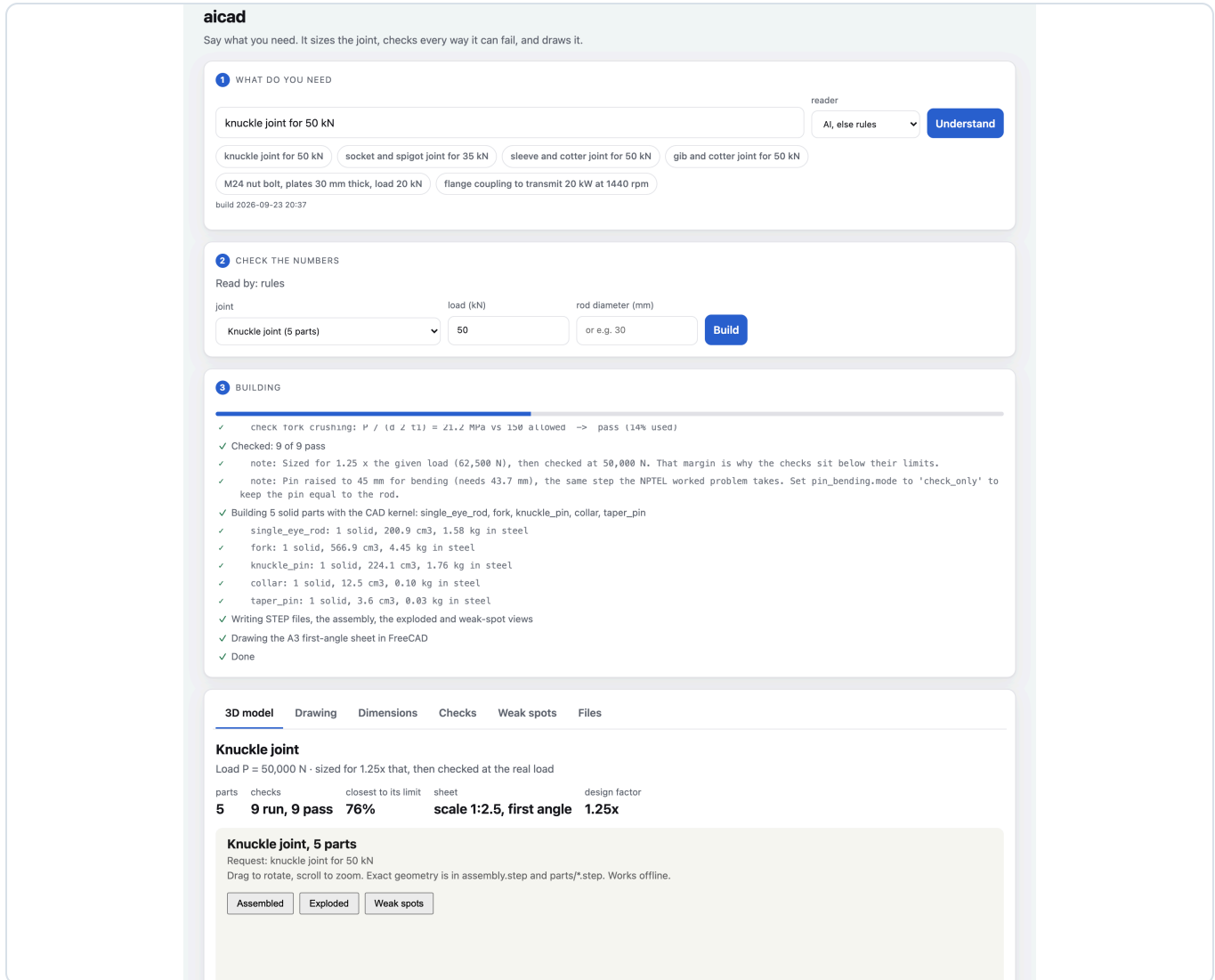
WORKED EXAMPLE · 3 OF 3

The working is printed, so the output can be checked rather than trusted

Each run prints the values computed before rounding, the size adopted, the rule and source behind every proportion, every check with its arithmetic substituted, and the volume and mass of each solid produced.

> **Why this matters for a review.** A reviewer does not have to accept the result. Each line can be recomputed by hand from the formula shown beside it.

This is also what made the errors in the published sources visible, on the following slides.



Working printed for a single run of the example above.

OUTPUT · GEOMETRY

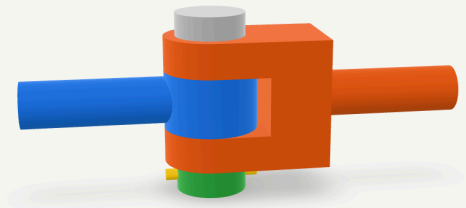
Exact solids, assembled, exploded, and coloured by utilisation

Knuckle joint, 5 parts

Request: knuckle joint for 50 kN

Drag to rotate, scroll to zoom. Exact geometry is in assembly.step and parts/*.step. Works offline.

Assembled | Exploded | Weak spots



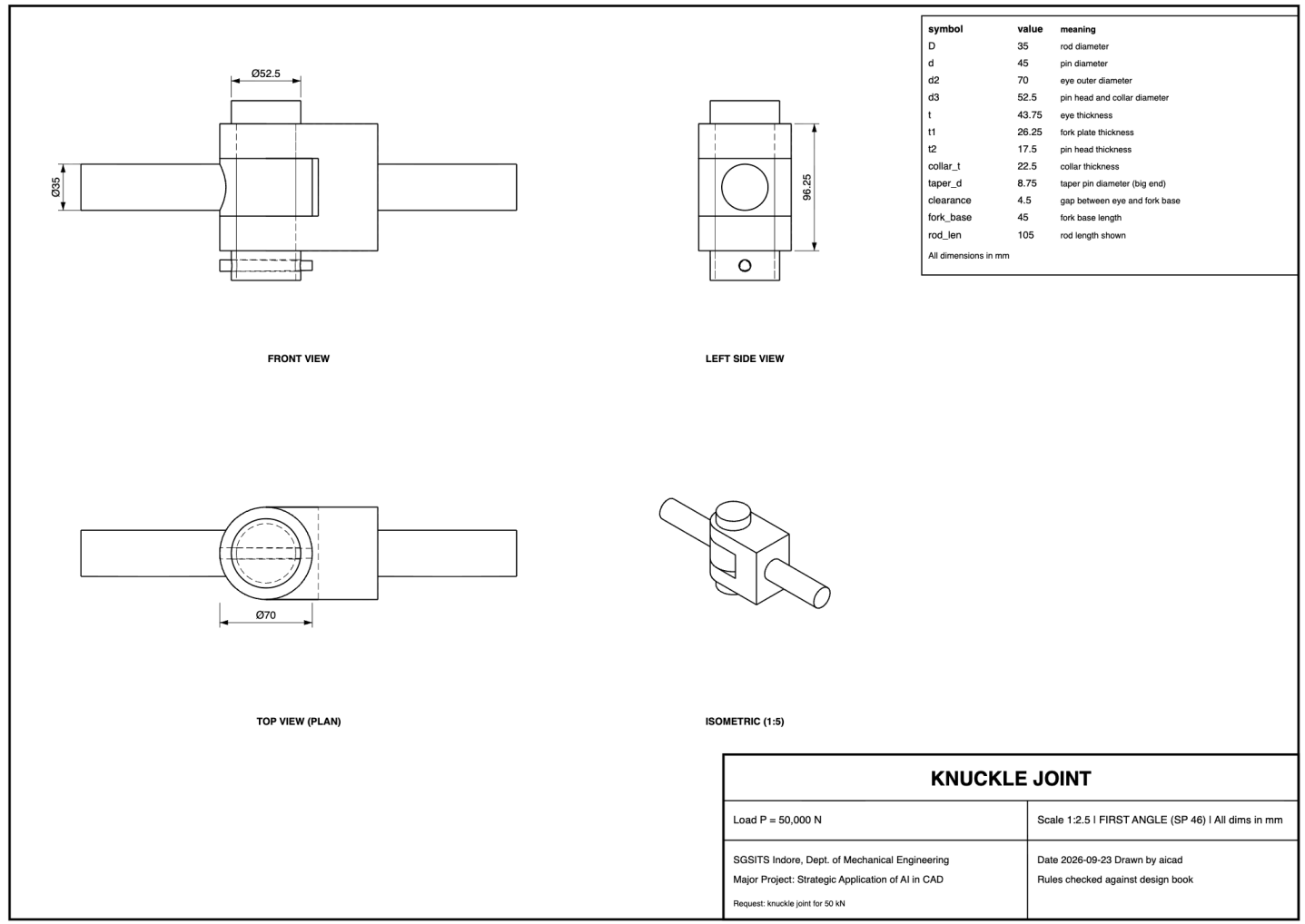
Assembled model, rotatable in the browser.

FILE	USE
`parts/*.step`	each component separately
`assembly.step`	all parts in position
`drawing.svg`	A3 first angle sheet
`drawing.dxf`	same views for AutoCAD
`*.FCStd`	editable FreeCAD file
`report.md`	dimensions, rules and checks

> **STEP is a neutral format.** The same file opens in Siemens NX, SolidWorks, FreeCAD, Onshape and Fusion, so the output is not tied to this software.

OUTPUT · DRAFTING

First angle A3 sheet, generated without opening a CAD application



Front view, left side view, plan and isometric, with dimension lines and a title block carrying the load and the request. Produced by driving FreeCAD without its graphical interface, projecting the assembled geometry and composing the sheet.

VERIFICATION

Checked against published design literature

JOINT	SOURCE CONSULTED	OUTCOME
Knuckle joint	NPTEL IIT Kharagpur, Module 4 Lesson 2	proportions and all nine checks verified
Socket and spigot cotter	NPTEL M4 L2; Leutwiler (1917) Art. 98	all ten checks verified against two independent books
Gib and cotter	Unwin (1899), p. 179, Figs. 130 and 131	breadth split, thickness and taper verified
Flange coupling	NPTEL M4 L1 keys; Leutwiler Art. 275 bolts	all seven checks verified twice; proportions cannot be
Sleeve and cotter	four books searched, none covers it	checks verified by correspondence; proportions not
Hex bolt and nut	NPTEL Module 4 Lesson 4	tightening relation verified; size table not

Where a source does not exist, the software says so on every run. Leutwiler records that no standard set of flange coupling proportions was ever agreed and that published tables are individual manufacturers' practice. Those proportions are therefore marked unverified rather than given a source they do not have.

FINDING 1 OF 3

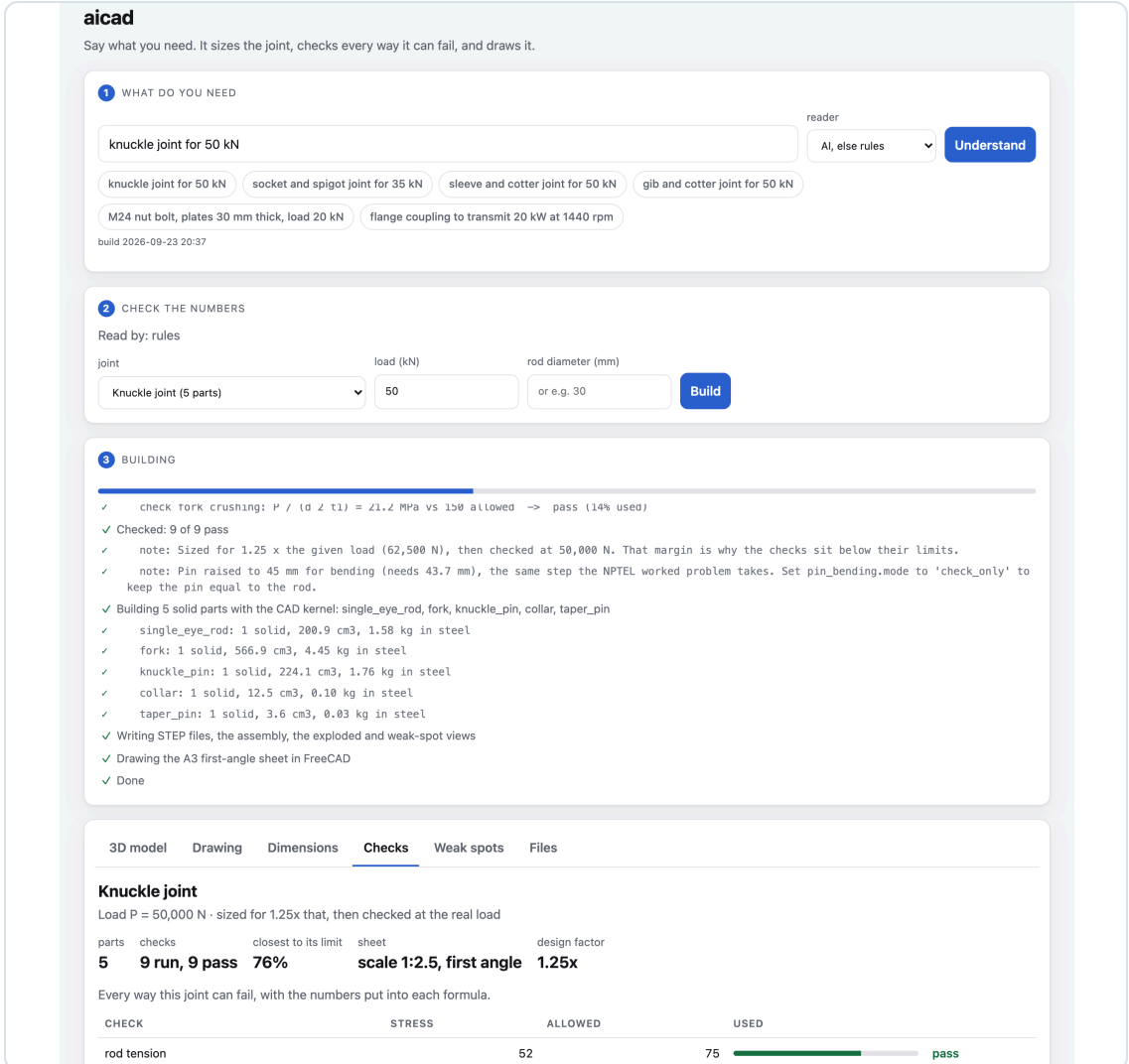
A worked example that fails its own check

The knuckle joint worked problem raises the pin from **40 mm to 55 mm** to satisfy bending. It then evaluates the eye using the **original 40 mm pin**.

```
eye shear = P / ((d₂ − d) t)
with d = 55: 100000 / ((80 − 55) × 50) = 80 MPa
allowable = 65 MPa
```

> **Its own expression, evaluated with the pin it actually adopted, exceeds its own allowable by 23 percent.**

What this software does instead: it enlarges the outer diameter of the eye until the check passes, and reports that it has done so.



Because every check is evaluated and its arithmetic printed, an inconsistency of this kind becomes visible immediately rather than being inherited.

FINDINGS 2 AND 3 OF 3

An omitted failure mode, and a constant printed wrongly

Omitted failure mode

The cotter joint procedure verifies crushing of the socket but not crushing between the cotter and the spigot.

```
P / (d₁ t) = 50000 / (30 × 10) = 167 MPa
allowable = 110 MPa
```

Leutwiler, Article 98, lists ten failure modes and includes this one. Two independent texts, published eighteen years apart, contain a check the lesson omits. That is what makes it an omission rather than a preference. The software keeps the check and sizes the cotter thicker.

Constant printed wrongly

The initial tightening load for a bolt is printed as 284*d* kN. The worked example on the same page obtains 56.8 kN for an M20 bolt.

```
284 × 20 = 5.68 kN   ≠   56.8 kN
2840 × 20 = 56,800 N   =   56.8 kN ✓
```

The page refutes itself. The constant is 2840*d* newtons, and the printed relation is out by a factor of ten. Every bolted design now reports the tightening load, because at M20 it is 56,800 N before any service load is applied.

INDEPENDENT CROSS CHECK

Finite element analysis, and knowing when not to believe it

CASE	FORMULA	COMPUTED	DIFFERENCE
20 mm bar, 10 kN tension	31.83	32.22	+1.2 %
Knuckle eye rod, 50 kN	52.0	57.9	+11.5 %

The bar validates the method. The eye rod reads higher because the formula treats it as a plain bar and takes no account of the stress raised at the pin bore. Both remain below the 75 MPa allowable.

A result that never settles is not a stress. On one component the computed peak moved between **157 and 191 MPa** across a sixteen fold increase in nodes. That is a singularity at a sharp corner or a constrained face, not a value to design to.

The software therefore re-solves on a mesh of half the element size whenever the peak exceeds 1.5 times the calculated value, and reports `NOT CONVERGED` if the peak then moves more than 5 percent.

Stated plainly in the report: this is one linear static run on one component with boundary conditions the software chooses. It is a second opinion on the formula, not a substitute for a full analysis.

CORRECTNESS

290 automated tests, including tests that protect the findings

WHAT IS TESTED	WHY
Sizing arithmetic	values match the worked problems in the sources
Generated geometry	each part is one valid solid, no two parts overlap
Request interpretation	parameters extracted correctly, invented numbers dropped
Drawing sheet	sheet, DXF and FreeCAD file are produced
Finite element path	face selection refers to dimensions that exist
The three findings	assert the source errors are real

The last row is the unusual one. Tests assert that the published answers fail their own checks. Without them, someone could later "correct" the software back to the printed values and every other test would still pass.

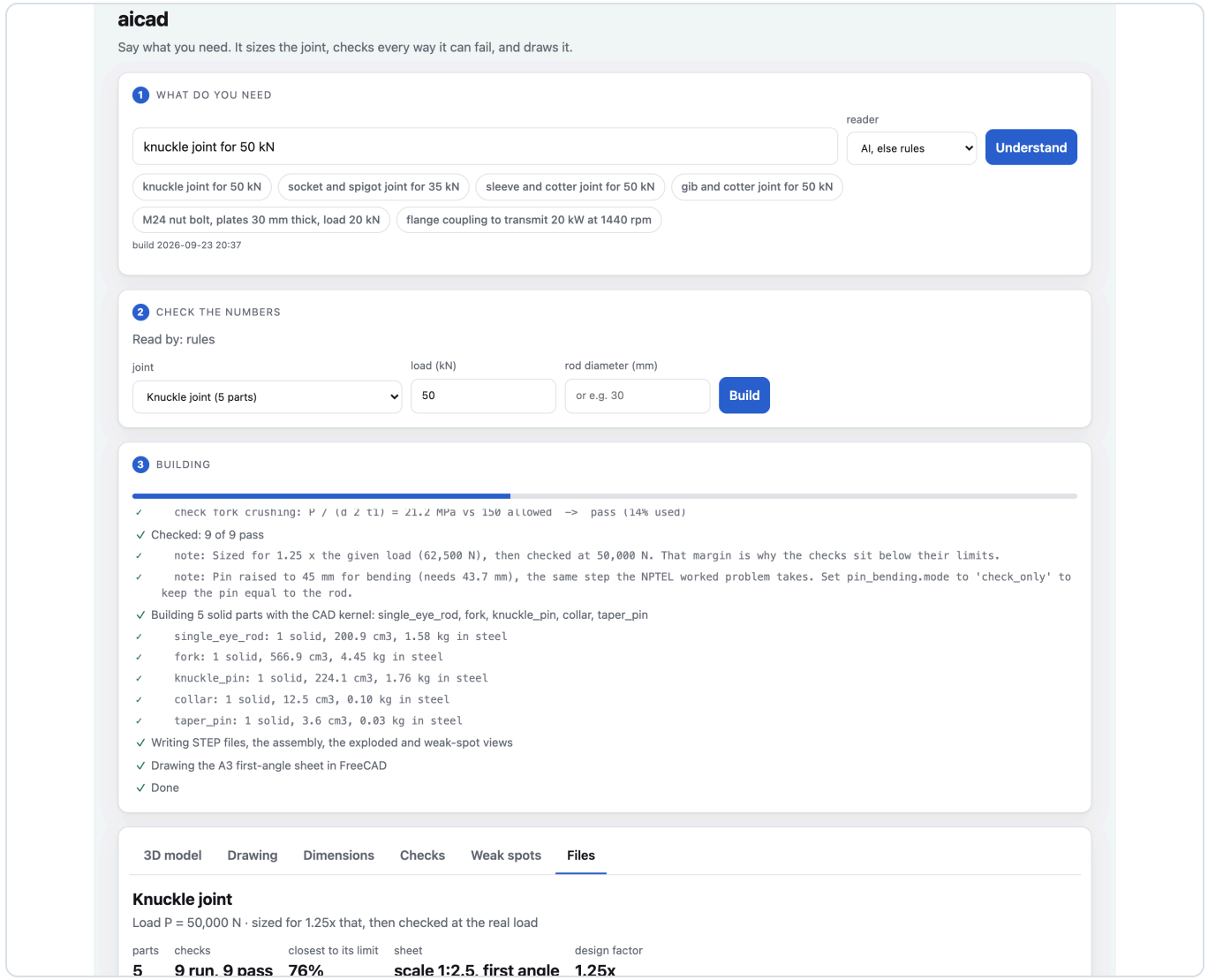
Overlap testing found a real error. A gib hook was modelled 1.25 mm inside the strap, occupying metal that another part already occupied. No dimension check would have caught it; only comparing the solids did.

DEPLOYMENT

Runs in a browser, with nothing installed

COMPONENT	TOOL
Solid modelling	CadQuery 2.8, OCCT kernel
Drawing and DXF	FreeCAD, headless
Finite element	CalculiX with gmsh
Request reading	Claude, rules as fallback
Application	Python standard library
Hosting	container, free tier

Measured requirement: the CAD kernel alone needs 449 MB of memory and a full run peaks near 800 MB, which is what determined the hosting choice.

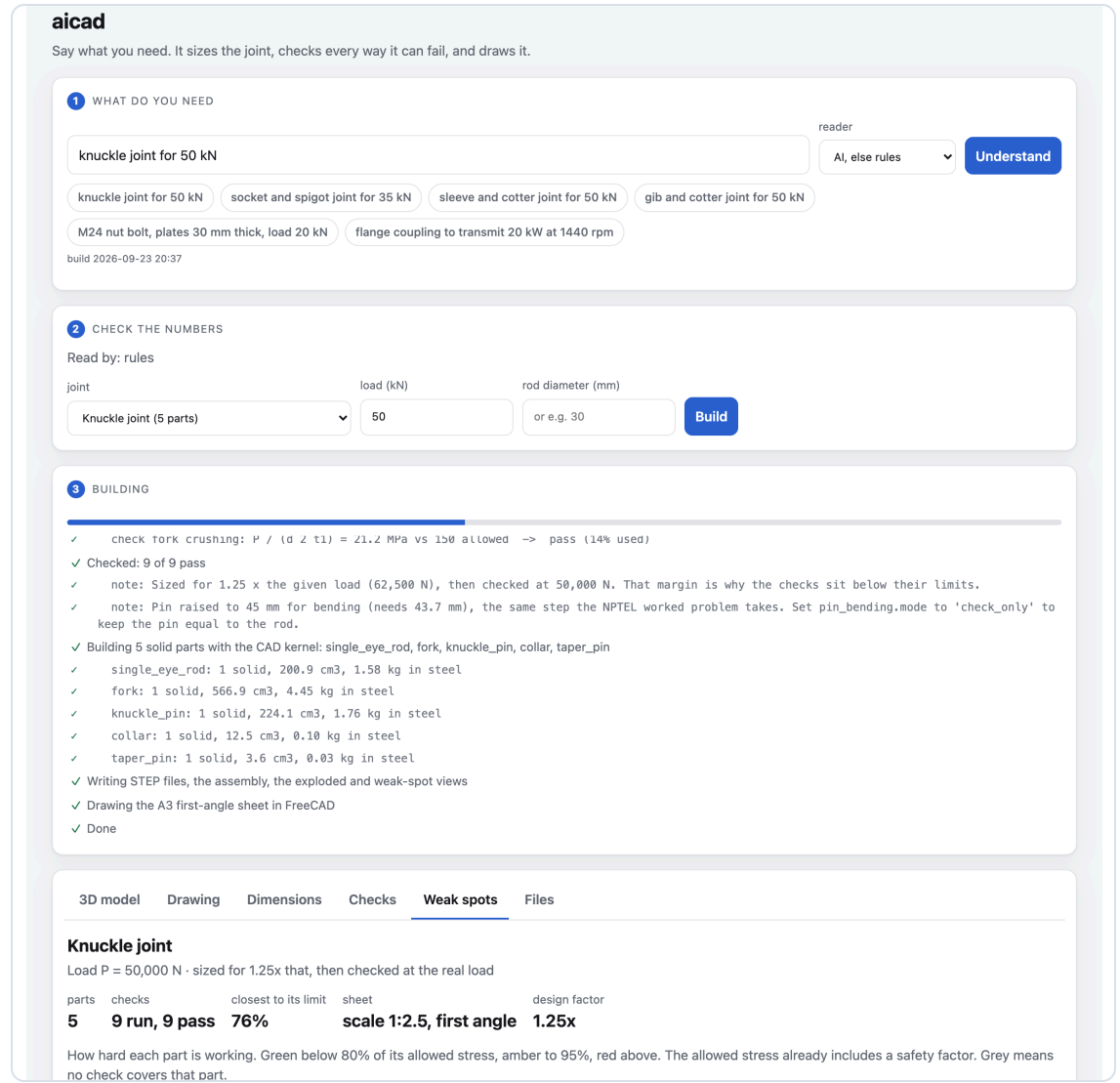


Files produced by one run, downloadable directly from the browser.

LIMITS

What this does not do, stated before anyone asks

- **Static loading only.** Fatigue, endurance limit and stress concentration factors are outside the scope.
- **Nominal dimensions only.** Tolerances and fits are not applied, so the sheet is not yet a manufacturing drawing.
- **Two joints have unverified proportions.** The sleeve and cotter joint and the flange coupling. The software reports this on every run.
- **Nut thread stripping is not checked** for the bolted joint.
- **The published site reads with pattern rules**, not the language model, because the server holds no API key.



Components coloured by utilisation, so the member nearest its limit is visible without reading the table.

FUTURE SCOPE

Where this goes next

- **Riveted and welded joints.** Treated in Module 10 of the same source and following the same pattern, so they need no change to the architecture.
- **Fatigue loading.** An endurance limit and a stress concentration factor would extend the system to cyclically loaded joints.
- **Tolerances and fits** for the pin and its bore, which would make the sheet directly usable for manufacture.
- **Verification of the two remaining proportion sets** from a design data handbook.
- **A journal script for Siemens NX**, so the assembly is constructed inside the software rather than imported into it.
- **Bill of materials and cost**, extending the material selection that is already partly implemented.

CONCLUSION

Every dimension it produces can be traced to a formula, and every formula to a page in a book.

The language model reads the request. The engineering stays deterministic, checked and open to inspection. That division is what allows a natural language interface without any part of the design becoming unverifiable.

6 joints, 44 checks
every one verified against the allowable stress on every run

3 errors found
in the published sources, protected by tests so they cannot be silently reverted

290 tests
covering arithmetic, geometry, parsing, drafting and analysis

aicad-vercel.vercel.app